



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
PRE-SCORE
17-11-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : I(A), TTAS & TEAS)

PAPER-1

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS
SECTION-I (i) : (Maximum Marks: 18)

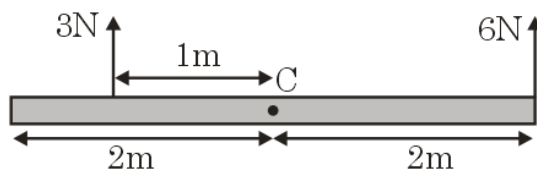
- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

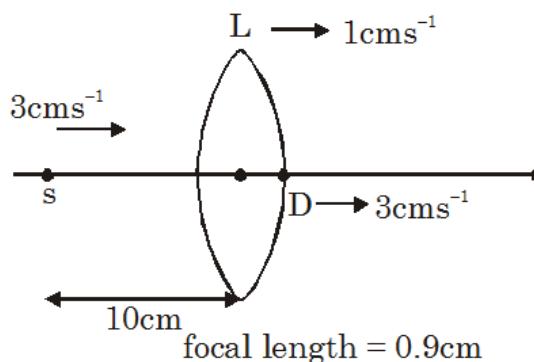
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A rod of mass 1 kg is kept on frictionless table and is acted upon by 2 forces as shown. How far from the center C of the rod should we apply a single force which produces the same effect as produced by the two forces shown?



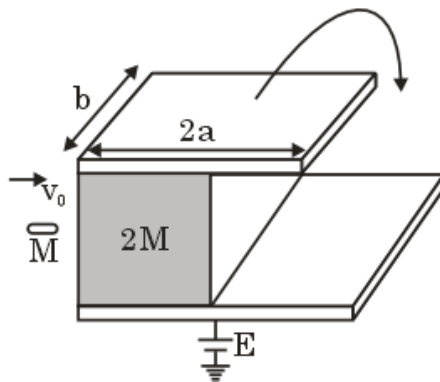
- (A) 1m (B) 1.1m (C) 1.2m (D) 1.5m
2. The figure shows the initial position of a point source of light s, a detector D and lens L. Now at $t = 0$ all three starts moving towards right with different velocity as shown in figure. The times at which detector receives the maximum light. Assume that the detector is initially just touching the lens.



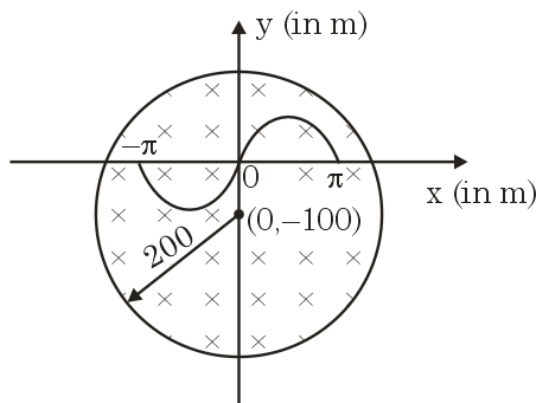
- (A) 5 s and 10 s (B) 0.5 s and 4.5 s (C) 2 s and 3 s (D) Only 10 s

Space for Rough Work

3. A parallel plate capacitor is half filled with a dielectric (K) of mass $2M$. Capacitor is attached with a cell of emf E . Plates are held fixed on smooth insulating horizontal surface. A bullet of mass M hits the dielectric elastically and it is found that dielectric just leaves out the capacitor. Find speed of bullet



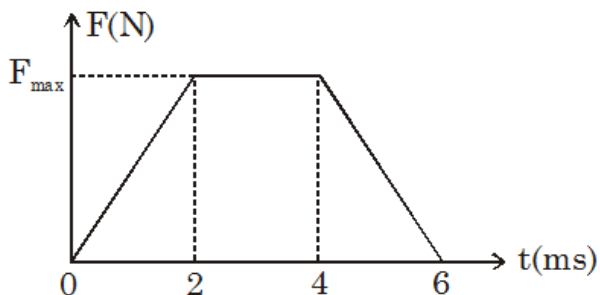
- (A) $E \left[\frac{\epsilon_0 ab (K - 1)}{Md} \right]^{1/2}$ (B) $E \left[\frac{\epsilon_0 ab (K - 1)}{2Md} \right]^{1/2} \times \frac{2}{3}$
- (C) $E \left[\frac{\epsilon_0 ab (K - 1)}{Md} \right]^{1/2} \times \frac{3}{2}$ (D) $E \left[\frac{\epsilon_0 ab (K - 1)}{2Md} \right]^{1/2} \times \frac{3}{2}$
4. A time varying uniform magnetic field, varying at constant rate 1 T/sec exists in a circular region of radius 200 m centered at $(0, -100)$. A conducting wire is placed along $y = \sin kx$, where $k = 1 \text{ rad/m}$, from $x = -\pi$ to $+\pi$. Find the magnitude of e.m.f. generated in the wire.



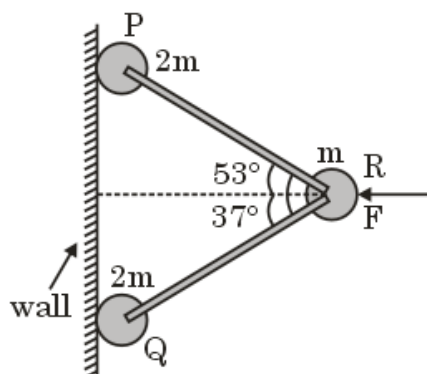
- (A) 157 V (B) 314 V (C) 628 V (D) Zero

Space for Rough Work

5. Figure shows F vs t graph during collision of a 58-g tennis ball with a wall. The ball moves perpendicular to the wall with speed of the 32 m/s & rebounds perpendicularly with same speed. What is the value of $F_{\max}$?



- (A) 715 N (B) 828 N (C) 878 N (D) 928 N
6. Three small discs are connected with two identical massless rods as shown in figure. The rods are pinned to the discs such that angle between them can change freely. The system is placed on a smooth horizontal surface with discs P and Q touching a smooth wall. A force F is applied to the disc R in direction perpendicular to the wall. Find acceleration of disc P immediately after the force starts to act. Masses of discs are $m_P = m_Q = 2m$, $m_R = m$ [wall is perpendicular to the plane of the figure]



- (A) $\frac{F}{\sqrt{2}m}$ (B) $\frac{\sqrt{2}F}{\sqrt{m}}$ (C) $\frac{2F}{11m}$ (D) $\frac{11F}{2m}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. Which of the following expressions represent SHM?

(A) $A \cos \omega t \sin^2 \omega t$

(B) $A \sin(2\omega t)$

(C) $A \sin \omega t + B \cos \omega t$

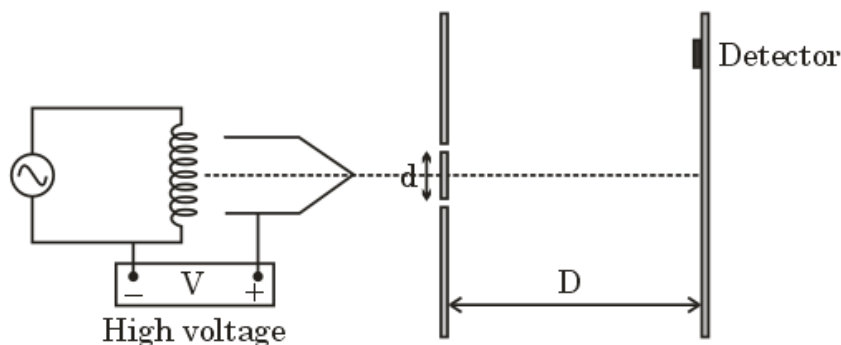
(D) $A \sin^2 \omega t$

Space for Rough Work

8. Mark the correct statements(s):
- (A) Direction of wave propagation is along the normal to wavefront
 - (B) For a point source of light, the shape of wavefronts can be plane at very large distance from the source
 - (C) A point source of light is placed at the focus of a thin spherical lens, then the shape of the wavefront for emerged light can be plane
 - (D) none
9. Emissivity of body depends on many which factors:
- (A) Temperature (B) Surface finish (C) Material (D) wavelength
10. When stopping potential is applied in an experiment on photoelectric effect, no photocurrent is observed. This means that
- (A) the emission of photoelectrons is stopped
 - (B) the photoelectrons are emitted but are reabsorbed by the emitter
 - (C) the photoelectrons are accumulated near the collector plate
 - (D) the photoelectrons are dispersed from the sides of the apparatus

Space for Rough Work

11. As shown in the figure; an electron gun emits thermal electrons with negligible speed. These electrons are accelerated through a potential difference V and made to pass through a hole. These electrons then pass through the slits on wall and finally hits a screen at different locations.



- (A) the wavelength of electrons after it comes out of the gun is $\frac{h}{\sqrt{2m eV}}$
- (B) fringe width can be increased by decreasing the voltage V .
- (C) If $V^2 > \frac{h^2}{2med^2}$, then at least 3 maxims will be obtained on the screen.
- (D) None of these
12. If the accelerating voltage across an x-ray tube is doubled
- (A) The wavelength of characteristic lines is halved.
- (B) The minimum wavelength of x-rays is halved
- (C) The x-rays probably become more penetrating
- (D) The target metal will get heat up earlier.

Space for Rough Work

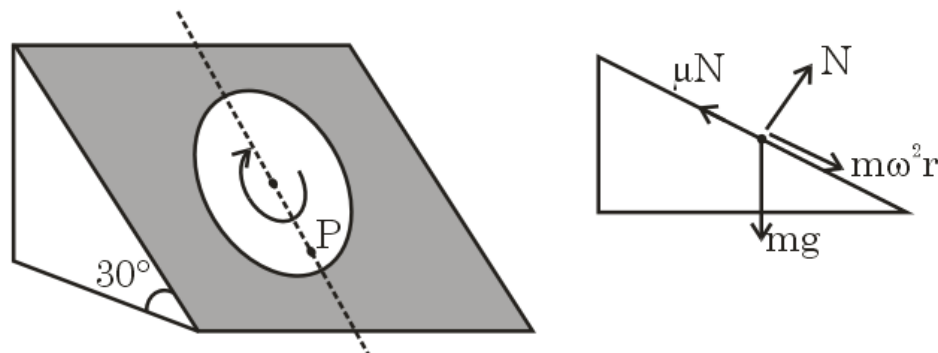
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
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1. An old record player of 10 cm radius turns at 10 rad/s while mounted on a 30° incline as shown in the figure. A particle of mass m can be placed anywhere on the rotating record. If the least possible coefficient of friction μ that must exist for no slipping to occur is μ , find $2\sqrt{3}\mu$.



2. An ideal monatomic gas is enclosed in a vertical athermic cylindrical container and supports a freely moving piston of mass m . The piston and the cylinder have equal cross-sectional area A . the piston is displaced from its equilibrium position. For small oscillations, find the angular frequency of piston (in radian/sec). Height of the gas in equilibrium position is h . Atmospheric pressure = 10^5 Pa. [Take $m = 10$ kg, $A = 10$ cm², $h = 5$ cm].

Space for Rough Work

3. An isolated copper ball of radius 9 mm having work function 4.47 eV is illuminated by electromagnetic radiation of wavelength 1243 Å. Find the maximum charge (in $\times 10^{-12}$ C) on the ball.
4. A hypothetical planet of mass M has three moons each of equal mass m , each revolving in the same circular orbit of radius R . The masses are equally spaced and thus form an equilateral triangle. Find the modulus of potential energy of system (in $\times 10^{26}$ Joules).
[Take $M = \sqrt{3}m = 3 \times 10^{22}$ kg, $G = 20/3 \times 10^{-11}$ m³ kg⁻¹ s⁻², $R = 400\,000$ km].
5. A visitor to a medieval castle measures the depth of a well by dropping a stone and timing its fall. She finds that the time to fall is $t = (3.0 \pm 0.5)$ sec and calculates the depth as $d = 0.5 \text{ gt}^2$. What is the relative error in calculation of 'd' if she takes $g = 9.80 \text{ m/s}^2$ with negligible uncertainty?
6. Ice at 0°C is added to 200 g of water initially at 70°C in a vacuum flask. When 50 g of ice has been added and all has melted; the temperature of the flask and contents is 40°C. When a further 80g of ice has been added and all has melted, the temperature of flask is 10°C. Latent heat of fusion of ice (in $\times 10^5$ J/ kg) is:

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 18)

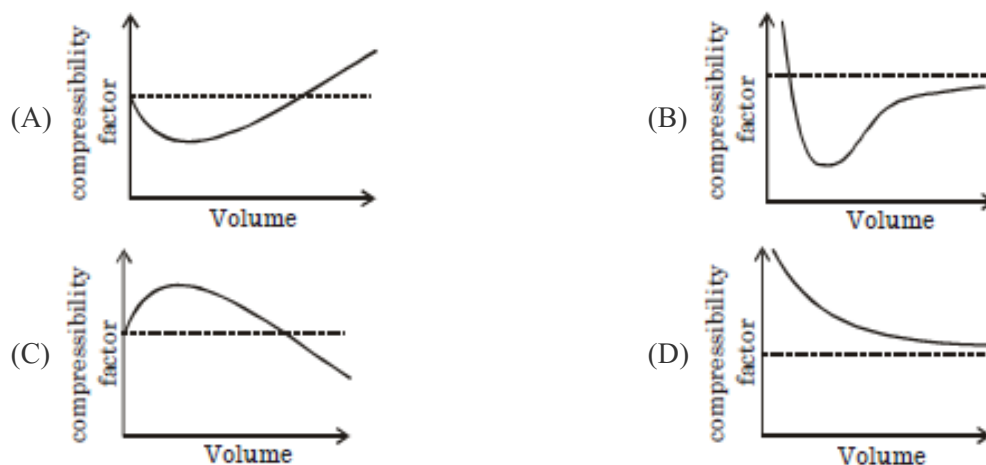
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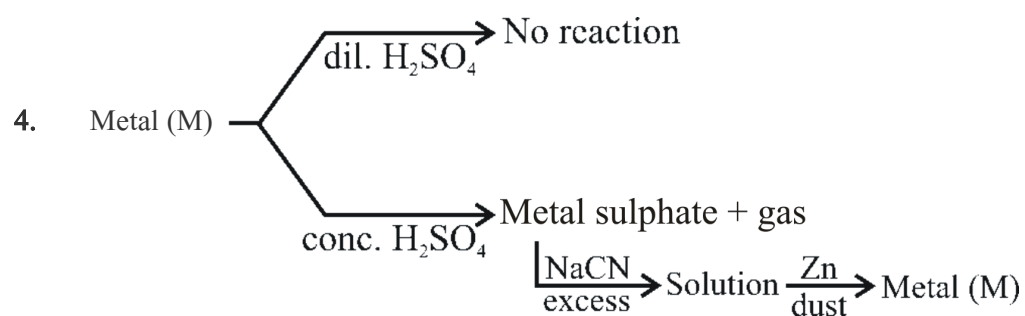
Negative Marks : -1 In all other cases

1. Select the correct curve for given mass of a real gas at constant temperature (dotted plot is for ideal gas) -



2. Given, $\text{H}_2(\text{g}) + \text{Br}_2(\text{g}) \rightarrow 2\text{HBr}(\text{g})$, ΔH_1^0 and standard enthalpy of condensation of bromine is ΔH_2^0 , standard enthalpy of formation of HBr at 25°C is
- (A) $\Delta H_1^0/2$ (B) $\Delta H_1^0/2 + \Delta H_2^0$
(C) $\Delta H_1^0/2 - \Delta H_2^0$ (D) $(\Delta H_1^0 - \Delta H_2^0)/2$
3. CuCl_2 is crystallised as emerald green colour having formula of $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$. This green colour is due to the presence of :
- (A) $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ complex (B) $[\text{CuCl}_4]^{2-}$ complex
(C) Presence of both (D) None of these

Space for Rough Work



Metal 'M' is

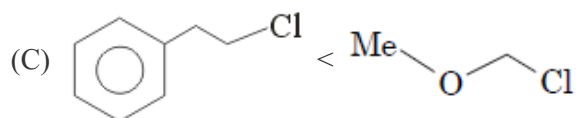
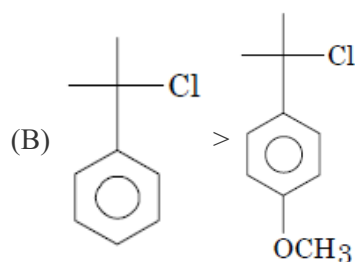
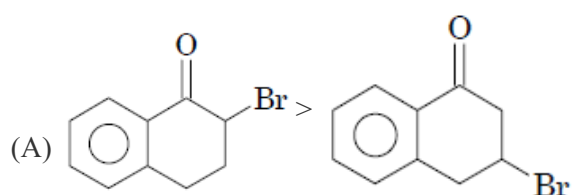
(A) Ag

(B) Au

(C) Fe

(D) Al

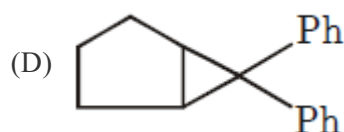
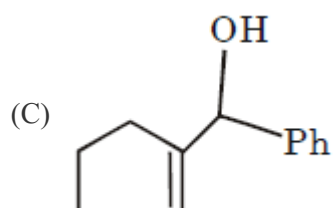
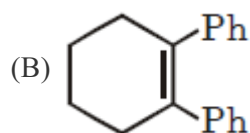
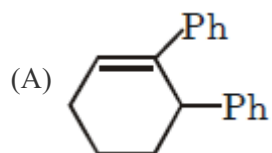
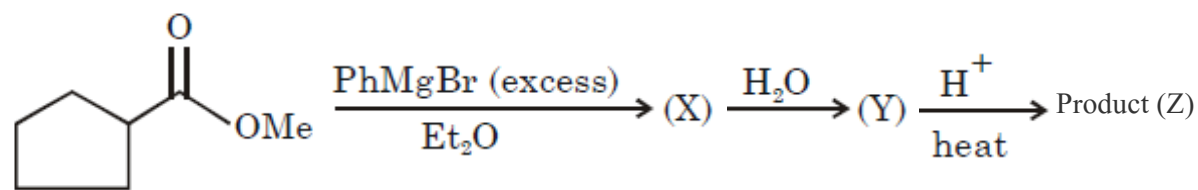
5. Correct order for S_N1 rate :



(D) All are correct

Space for Rough Work

6. Major product "Z" obtained in given reaction is ?



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

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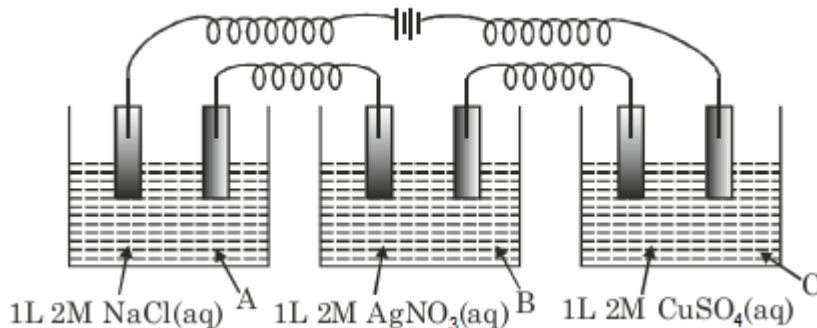
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7. Which of the following statements are true for hydrophilic sols?
- (A) They do not require electrolytes for stability
- (B) Their coagulation is reversible
- (C) Their viscosity is of the order of water
- (D) Their surface tension is usually lower than that of dispersion medium

Space for Rough Work

8. Three electrolytic cell with inert electrodes are connected in series combination as following :
If 9.65 amp current is passed from external battery for 100 second then which is / are correct statements?

(Assume volume of solution in electrolytic cells remain constant during electrolysis) (Atomic mass of Ag = 108)



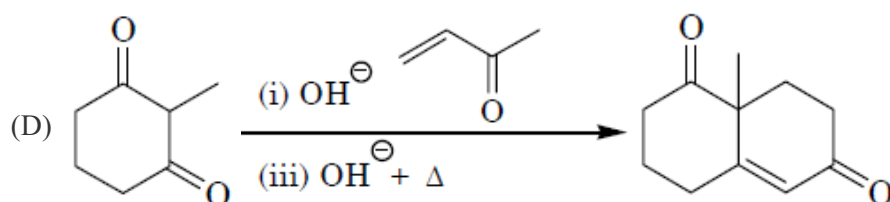
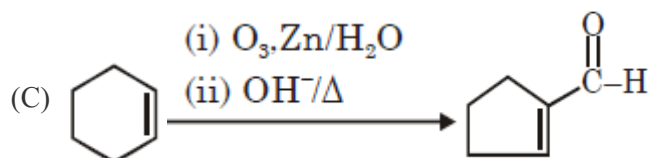
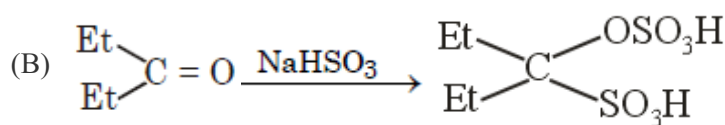
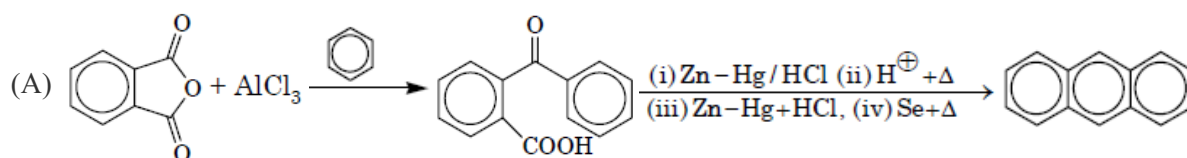
- (A) pH of electrolytic cell A is 12 after electrolysis.
(B) 21.6 gm of silver deposits at cathode in electrolytic cell B after electrolysis.
(C) Blue colour intensity in the solution of electrolyte cell C, decreases during electrolysis.
(D) Electrolytic cell B and electrolytic cell C have equal pH after electrolysis, assuming no hydrolysis of both salts.
9. Which of the following substance(s) produce Brown colouration with solution of NaI in dilute CH₃COOH :-
- (A) CaOCl₂
(B) CuSO₄
(C) CaCl₂
(D) Gas (X) which is obtain on anode by electrolysis of NaCl_(aq)

Space for Rough Work

10. Which of the following statement(s) is/are correct for extraction of Aluminium?

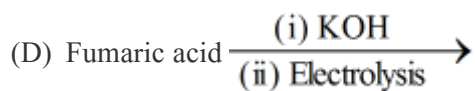
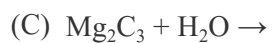
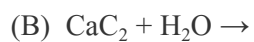
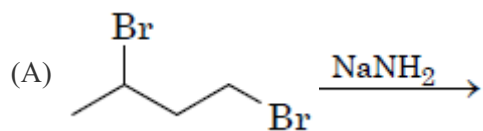
- (A) In electrolytic refining of Al, BaF_2 is used to decrease m.p. of Al
 (B) White bauxite is concentrated by Serpeck's process.
 (C) CO_2 & CO are produced at anode in electrolytic reduction of Al_2O_3
 (D) Electrolytic reduction of Al_2O_3 gets completed in Hall-Heroult cell.

11. Which of the following reaction(s) give correct product as shown



Space for Rough Work

12. In which of the following reaction(s), an alkyne product can be formed ?



Space for Rough Work

SECTION-II : (Maximum Marks: 24)

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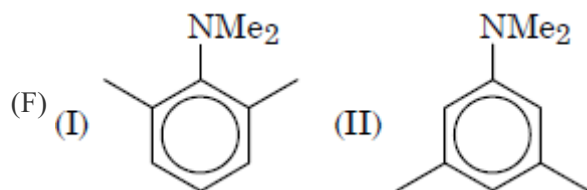
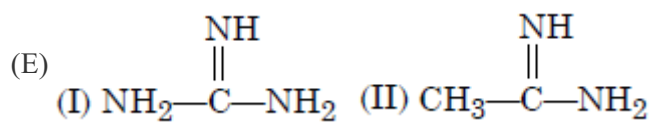
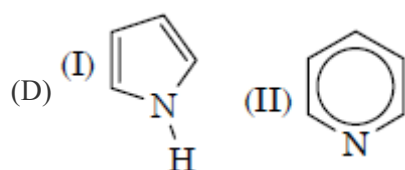
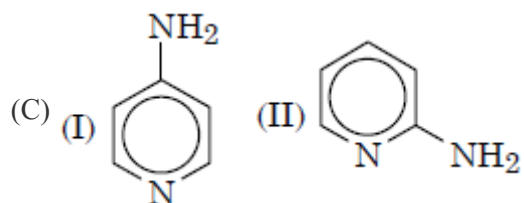
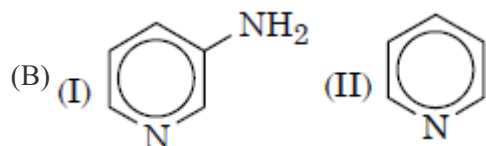
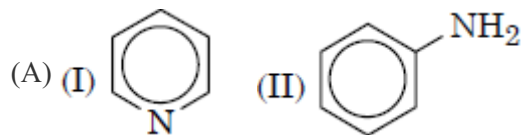
Zero Marks : 0 In all other cases.

1. For the first order reaction, $A(g) \rightarrow 2B(g) + C(g)$ the half life for decomposition of A is 3 minutes at 300K. Calculate the time (in minute) in which partial pressure of A(g) will drop from 2 bar to 0.5 bar at 400K. Given activation energy of reaction is 840 R. (Take $\ln 2 = 0.7$)
2. Solid $KClO_3$ is taken in a container maintained at constant pressure of 1 atm. Upon heating following equilibria is set :

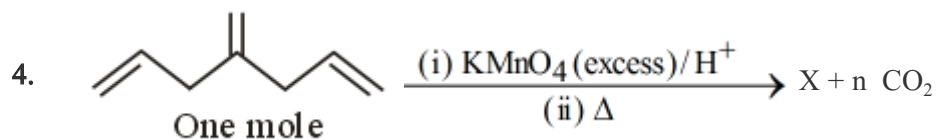
$$2KClO_3(s) \rightleftharpoons 2KCl(s) + 3O_2(g)$$
 If $\Delta H^\circ = 25 \text{ kcal/mol}$ and $\Delta S^\circ = 50 \text{ cal/K}$, at what temperature (in K) equilibrium will be established in the container.
 (Ignore variation of ΔH° and ΔS° with temperature.)

Space for Rough Work

3. Out of following in how many cases Ist is more basic than IInd.



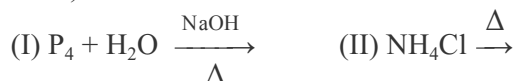
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Value of 'n' for complete reaction?

5. Total number of complex compound which have low spin as well as Diamagnetic.
 $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{NiF}_6]^{2-}$, $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Fe}(\text{CN})_6]^{4-}$, $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$, $[\text{Ni}(\text{en})_3]^{2+}$,
 $[\text{PtCl}_4]^{2-}$

6. Find the number of reaction in which XH_3 is obtained as one of the gaseous product (where X is central atom)



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PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- Each question has Four options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Number of pairs (x, y) of real numbers that satisfy the equation

$$\left(\sin^2 x + \frac{1}{\sin^2 x}\right)^2 + \left(\cos^2 x + \frac{1}{\cos^2 x}\right)^2 = 12 + \frac{1}{2}\sin y$$

(A) 0 (B) 1 (C) 2 (D) Infinite
2. Let $\int_0^a \{x\} dx = \frac{41}{8}$ and $\int_0^{b\pi} |\cos x| dx = 11$ (where a and b are positive real numbers), then value of a + b is
 (where {·} denotes fractional part function)-

(A) 16 (B) 18 (C) 10 (D) 8
3. The area of the region which consists of all the points satisfying the conditions $|x - y| + |x + y| \leq 8$, $xy \geq 2$, $x \geq 0$ and $y \geq 0$ is equal to -

(A) $14 - \ln 64$ (B) $28 - \ln 64$ (C) $14 + \ln 4$ (D) $18 - \ln 4$

Space for Rough Work

4. Let $g : \left[\frac{1}{2}, 1\right] \rightarrow (0, \infty)$, $g(x) = e^{-2x}f(x)$, where $g(x)$ is non constant decreasing and differentiable function and $g\left(\frac{1}{2}\right) = \frac{1}{e}$. Then the value of $\left[\int_{\frac{1}{2}}^1 f(x) dx\right]$ is
[where $[x]$ denotes the greatest integer less than or equal to real number x]
- (A) 0 (B) 1
(C) 2 (D) cannot be determined
5. Number of values of x satisfying the equation $\cos^{-1}(x^2 - 5x + 6) - \cos^{-1}x = \sin^{-1}x$, is/are
(A) 2 (B) 1 (C) 0 (D) infinite
6. If $1, \alpha_1, \alpha_2, \alpha_3, \alpha_4$ are roots of equation $x^5 - 1 = 0$, then the value of $\prod_{i=1}^4 \frac{\omega - \alpha_i}{\omega^2 - \alpha_i}$ is (where ω is imaginary cube root of unity)
(A) 1 (B) ω (C) ω^2 (D) 0

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
7. If $f(x)$ be twice derivable function such that $f''(x)$ is positive in $[0,4]$, then which of the following is(are) always CORRECT ?
- (A) $f(0) + f(4) = 2f(2)$ for atleast one $c \in (0,4)$
- (B) $f(0) + f(4) < 2f(2)$
- (C) $f(0) + f(4) > 2f(2)$
- (D) $2f(0) + f(4) > 3f\left(\frac{4}{3}\right)$
-

Space for Rough Work

8. If $A = \begin{pmatrix} 2 & 0 & 1 \\ 3 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$, then which of the following is(are) INCORRECT

(where A^{-1} is inverse of matrix A, I is an unit matrix of order 3, adj A is adjoint of matrix A, $|A|$ denotes determinant value of matrix A.)

(A) $\frac{1}{81} |\text{adj}(2A) \cdot \text{adj}(3A^{-1}) \cdot A^{-1}|$ is equal to 288

(B) $(A^{-1})^2 = \frac{1}{2}(A - 5I + 7A^{-1})$

(C) $|A^5 - 2A^4|$ is equal to 16

(D) $A^{-1} = \frac{1}{2}(A^2 - 5A + 6I)$

9. A bag contains n white and n black balls. Pairs of balls are drawn from the bag without replacement until the bag is empty. Probability that each draw consists of one white and one black ball is given by P(n), then-

(A) $P(2) = \frac{2}{3}$

(B) $P(6) = \frac{32}{231}$

(C) $P(2) = \frac{1}{3}$

(D) $P(6) = \frac{16}{231}$

10. Let $f(x) = |5 \sin x + 12 \cos x| ((x + m)^2 - n\pi(x + m) + p\pi^2)$ where $p, m, n \in \mathbb{R}$ and $m > 0$ is derivable in $(0, 2\pi)$, then

(A) $n + 2p = 6$

(B) $m = \sin^{-1} \left(\frac{12}{13} \right)$

(C) $p^2 + n = 7$

(D) $p^3 - 3n + 1 = 0$

Space for Rough Work

11. Let the angle between vectors $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{6}$, between vectors $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{4}$ and between vectors $\vec{c}$ and $\vec{a}$ is $\frac{\pi}{3}$. The angle, the vector $\vec{a}$ makes with the plane containing vectors $\vec{b}$ and $\vec{c}$, is

(A) $\cos^{-1} \sqrt{1 - \sqrt{\frac{2}{3}}}$

(B) $\cos^{-1} \sqrt{2 - \sqrt{\frac{3}{2}}}$

(C) $\sin^{-1} \sqrt{\sqrt{\frac{3}{2}} - 1}$

(D) $\sin^{-1} \sqrt{\sqrt{\frac{2}{3}}}$

12. Let $L_1 : \vec{r} = (\hat{i} - \hat{j} - 10\hat{k}) + \lambda (2\hat{i} - 3\hat{j} + 8\hat{k})$ and $L_2 : \vec{r} = (4\hat{i} - 3\hat{j} - \hat{k}) + \mu (\hat{i} - 4\hat{j} + 7\hat{k})$ represent two lines in $\mathbb{R}^3$, then which of the following is/are true

(A) L_1 is parallel to $4\hat{i} - 6\hat{j} + 16\hat{k}$

(B) L_2 is parallel to $-\hat{i} + 4\hat{j} - 7\hat{k}$

(C) L_1 and L_2 are coplanar

(D) Angle between the lines L_1 and L_2 is $\cos^{-1} \left(\frac{70}{11\sqrt{7}} \right)$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A circle $S = 0$ passes through points of intersection of circles $x^2 + y^2 - 2x + 4y - 1 = 0$ and $x^2 + y^2 + 4x - 2y - 5 = 0$ and cuts the circle $x^2 + y^2 = 4$ orthogonally. Then length of tangent from origin on circle $S = 0$ is
2. Let $A = \begin{bmatrix} \frac{1}{2} & \frac{3}{4} \\ 1 & -\frac{1}{2} \end{bmatrix}$ then the value of $\left(\frac{\text{tr}(A) + \text{tr}(A^2) + \text{tr}(A^3) + \dots + \text{tr}(A^{100})}{4} \right)^{\frac{1}{2}}$, is
3. Let (x, y) be a variable point on the curve $x^2 + 4y^2 - 4x + 8y + 4 = 0$.
Then $\frac{\max(x^2 + y^2 + 4x + 2y + 5)}{\min(x^2 + y^2 + 4x + 2y + 5)}$ is
4. A parabola passes through two points $A(1,0)$ and $B(-1,0)$ such that directrix always touches the $x^2 + y^2 = 4$ and if locus of focus of parabola is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, then $(a^2 + b^2)$ is
5. Let $(1 + x^2)^2(1 + x)^n = \sum_{k=0}^{n+4} a_k x^k$. If a_1, a_2 and a_3 are in Arithmetic progression then the sum of all possible value of n is :
6. Let $f(x)$ be a non constant thrice differentiable function defined on 'R' such that $f(x) = f(6 - x)$ and $f'(0) = 0 = f'(2) = f'(5)$. If n is the minimum number of roots of $f''(x)^2 + f'(x)f'''(x) = 0$ in $[0, 6]$ then value of $\frac{n}{2}$ is

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Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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